SYSTEMATIC EQUITIES / PARAMETER SENSITIVITY

# Rebalancing frequency and portfolio weights

## AAPL / TSLA: a two-dimensional study

QuantLab Research

20 October 2026

Frozen market sample: 6 January 2020–31 December 2025

PARAMETER GRID
**6,767**
67 schedules × 101 weights

WEEKLY 50/50 CAGR
**46.61%**
After modeled execution costs

WEEKLY 50/50 DRAWDOWN
**55.31%**
Maximum loss from a closing peak

**The weight decision dominates the broad risk profile.** More TSLA delivered greater historical growth and deeper losses. Rebalancing frequency changes the path and cost of that exposure, but the relationship is not monotonic: neither trading every day nor trading less often is uniformly best.

### Executive findings

The original weekly 50/50 strategy grows $100,000 to $990,605, with a 46.61% CAGR, 43.38% annualized volatility and a 1.102 Sharpe ratio. Its worst closing drawdown is 55.31%. These are exceptional historical returns accompanied by substantial equity risk.

At a fixed 50/50 weight, the recurring schedules span 43.47%–48.84% CAGR. At the weekly frequency, changing the AAPL weight from 0% to 100% spans 57.65%–25.12% CAGR and 73.63%–33.36% maximum drawdown. This comparison describes this sample; it is not a universal variance decomposition.

The full-sample maximum Sharpe occurs at **58% AAPL / 42% TSLA, every 17 sessions**: Sharpe 1.137, CAGR 45.92%, drawdown 52.02%. Its small Sharpe improvement over weekly 50/50 is an ex-post result from thousands of trials, without a significance claim.

A 2020–2022 training search selects a different Sharpe weight, 43% AAPL every 17 sessions. On a fresh 2023–2025 simulation, it has a lower Sharpe and a deeper drawdown than the prespecified weekly 50/50 reference, despite a slightly higher CAGR. The exact full-sample winner is therefore not established as a stable allocation rule.

**Scope.** Independent research note with a restrained institutional layout; no affiliation with Jump Trading, Citadel or any other investment firm. Results use archived Yahoo Finance data, not a live 2026 update. No leverage, shorting, tax model or external cash flows.

01 / RESEARCH DESIGN

# 1 Question, sample and execution

*Hold the investment universe fixed; vary only target weight and the predetermined rebalance calendar.*

| Item | Specification |
|---|---|
| Universe | AAPL and TSLA; fixed ex post |
| Weight grid | $w_{\text{AAPL}} = 0, 0.01, \ldots, 1$; $w_{\text{TSLA}} = 1 - w_{\text{AAPL}}$ |
| Session schedules | Every 1–63 trading sessions, anchored at first purchase |
| Calendar schedules | First session of each week, month and quarter |
| Reference | Buy once and hold for each of the 101 initial weights |
| Initial wealth | USD 100,000; cash baseline at 3 January 2020 close |
| Investment dates | 6 January 2020 open to 31 December 2025 close |
| Data observations | 1,507 sessions; 1,506 observed portfolio returns |
| Execution | Known calendar at prior close; next scheduled opening print |
| Costs | 5 bps adverse slippage; max(USD 1, USD 0.005 per unit) |
| Constraints | Long only; integer adjusted units; nonnegative cash |
| Final mark | Closing equity; no forced liquidation or exit cost |

### A calendar is known in advance; an opening price is not

The target weight and next rebalance date are fixed at the preceding session close. The broker sizes the already specified target at the next open using opening portfolio equity. It does not use that day's close to size or price an order. This is a target-weight execution convention, not a claim that a real market-on-open order can know the auction price in advance. There are no predictive signals or fitted factors.

For every-$k$-session schedules, investment-session index 0 is 6 January 2020 and the main grid rebalances at indices $0, k, 2k, \ldots$. A natural week is Monday–Sunday; holidays roll to the first available exchange session. A month or quarter uses the first exchange session of that period. Five sessions are not identical to one natural week; 21 sessions are not identical to one natural month.

### Adjusted research units

For each date, all OHLC prices are scaled by AdjClose/Close. This incorporates the provider's split and dividend adjustments; dividends and splits are not posted again. Holdings are integer *adjusted research units*, not historical legal share counts. This retains the original QuantLab convention, including per-unit commissions, but limits how literally fees and share quantities translate to a broker account.

Sales execute before purchases; AAPL precedes TSLA when cash constrains simultaneous buys. Slippage changes the target integer quantity as well as the fill price. The small resulting target mismatch is accepted; no residual order is retried between scheduled dates. Target weights drift with prices between rebalances.

**Controlled extension.** This study extends the archived weekly strategy without changing its costs or execution rules. Section 12 documents full-path replication and checks against the original broker.

02 / COMPLETE PARAMETER MAP

## 2 The joint return and risk surface

*Read vertically to change the weight; read horizontally to change the rebalance schedule.*

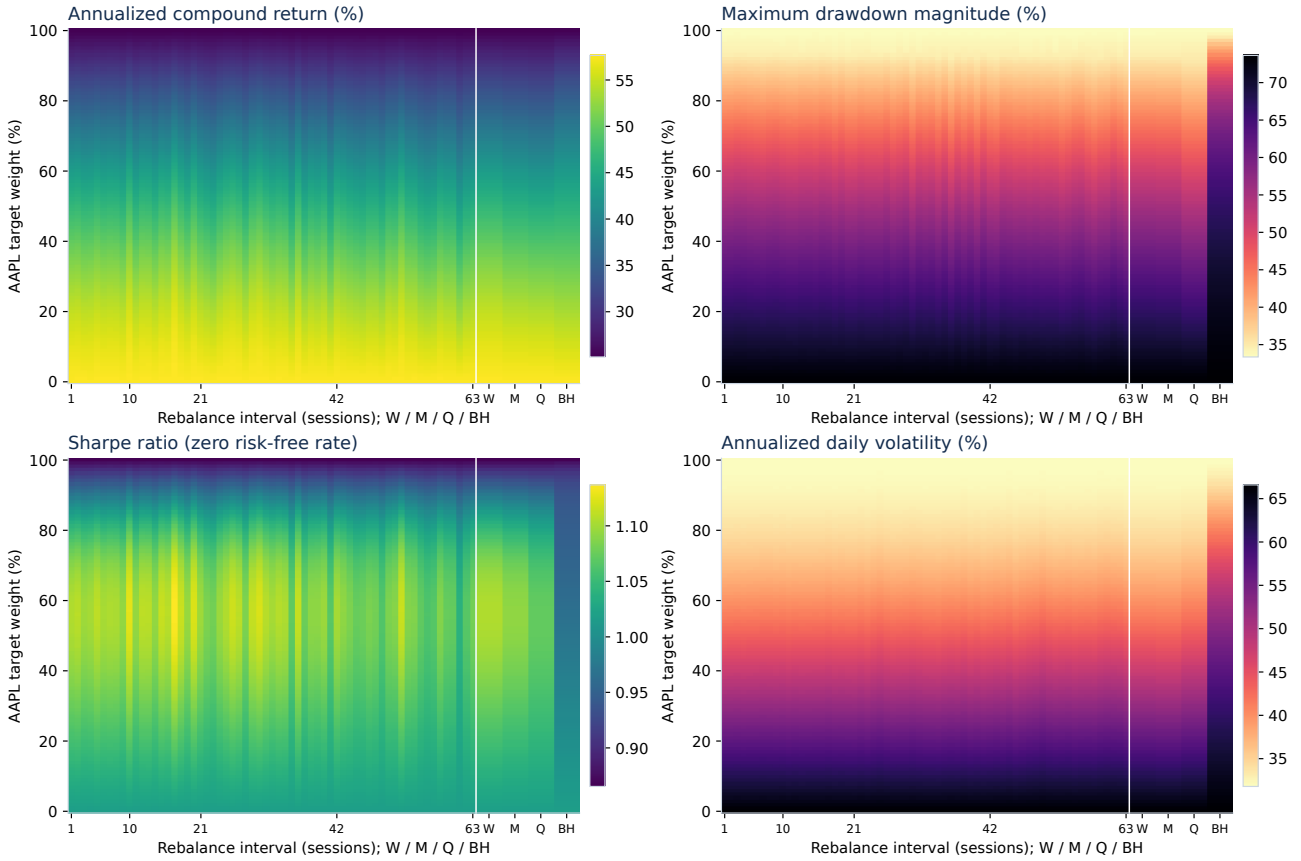



**Figure 1:** All 6,767 configurations under base costs. The first 63 columns are equally spaced session intervals; W, M, Q and BH are separate categorical schedules. BH denotes buy and hold. Calendar and buy-and-hold columns are widened for legibility; each panel has its own labeled color scale. Maximum drawdown is shown as a positive loss magnitude.

The weight axis produces the clearest broad gradient in both growth and risk. Higher TSLA exposure raises realized volatility and deepens the worst loss in this sample. The Sharpe surface is less monotonic: an interior blend can improve the ratio of mean return to daily volatility even when it sacrifices terminal wealth.

Across frequencies, the narrow bands and local ridges reflect the interaction between the calendar and the realized relative performance of the two stocks. They do not show a smooth law in which a longer interval must increase or decrease return. A single bright pixel is a historical maximum, not evidence that its exact interval is economically special.

The endpoint allocations are useful controls. At 100% AAPL or 100% TSLA, there is effectively no two-stock rebalancing to perform. In the base run, each endpoint makes only the initial purchase; all frequencies produce the same endpoint wealth to numerical precision.

**How to use this map.** First choose an acceptable exposure and loss profile. Then examine a neighborhood of frequencies at that exposure, together with costs and calendar sensitivity. The map alone cannot select an investable optimum.

03 / WEIGHT SENSITIVITY

## 3 The allocation sets the risk budget

*Higher return in this sample generally comes with a larger allocation to the more volatile stock.*

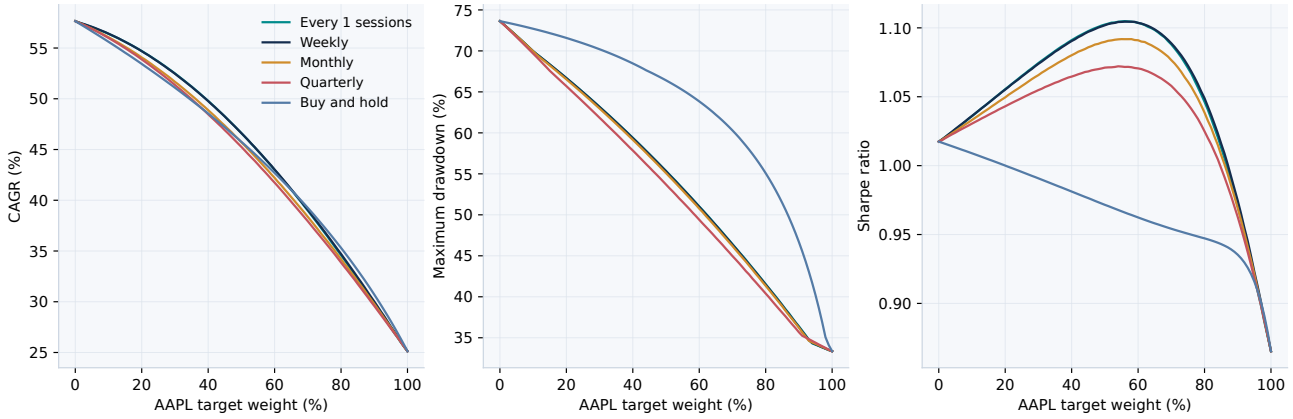



**Figure 2:** Weight slices for daily, weekly, monthly, quarterly and buy-and-hold schedules. The x-axis is the AAPL target weight; the TSLA target is its complement. For buy and hold, the weight is the initial allocation only.

| Weekly AAPL:TSLA | CAGR | Max DD | Volatility | Sharpe | Final wealth |
|---|---|---|---|---|---|
| 0:100 | 57.65% | 73.63% | 66.59% | 1.017 | $1,530,402 |
| 25:75 | 53.64% | 64.94% | 54.28% | 1.065 | $1,311,306 |
| 50:50 | 46.61% | 55.31% | 43.38% | 1.102 | $990,605 |
| 75:25 | 36.93% | 43.95% | 35.16% | 1.073 | $657,620 |
| 100:0 | 25.12% | 33.36% | 31.81% | 0.865 | $383,149 |

Moving from 50/50 to 75/25 AAPL/TSLA reduces the historical maximum drawdown from 55.31% to 43.95% and annualized volatility from 43.38% to 35.16%. The cost is a lower CAGR: 36.93% rather than 46.61%. This is the central portfolio tradeoff, not an execution improvement.

The weekly Sharpe maximum is at 56% AAPL, with a Sharpe of 1.104, compared with 1.102 at 50%. That difference is extremely small relative to the selection exercise and is not assigned statistical significance. A higher Sharpe also need not mean greater final wealth.

Buy and hold follows a different exposure rule after inception. The better-performing stock becomes a larger weight. Consequently, comparing a rebalanced 50/50 portfolio with an initially 50/50 buy-and-hold portfolio compares two different time paths of risk, not simply the presence or absence of transaction costs.

04 / FREQUENCY SENSITIVITY

# 4 Frequent trading does not guarantee more growth

*At a fixed allocation, calendar timing and trading costs compete with the effect of resetting weights.*

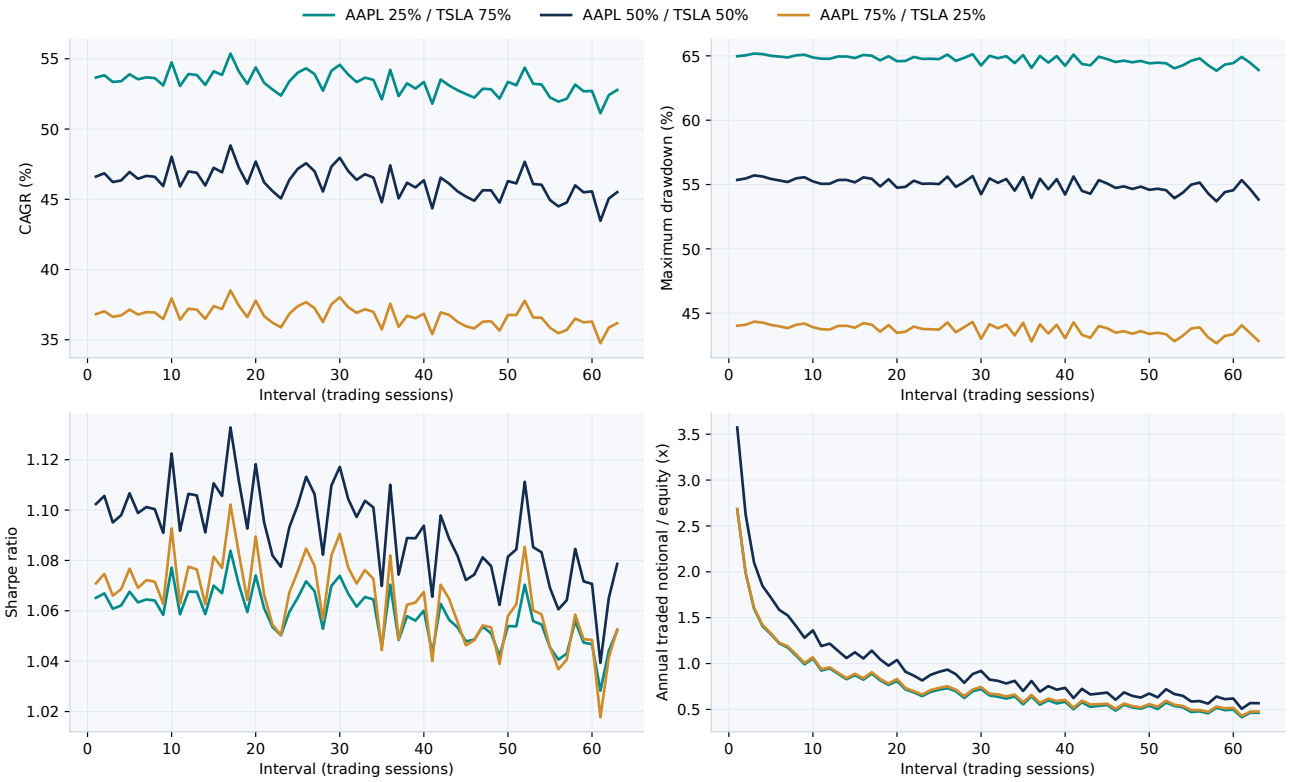



**Figure 3:** Session-interval slices at three fixed AAPL weights. Turnover is annual gross traded opening notional divided by pre-trade opening equity, including the initial purchase; it is not divided by two.

| 50/50 schedule | CAGR | Max DD | Sharpe | Turnover | Total cost |
|---|---|---|---|---|---|
| 1 sessions | 46.62% | 55.36% | 1.102 | 3.57x | $7,857 |
| 2 sessions | 46.85% | 55.47% | 1.106 | 2.62x | $5,096 |
| 3 sessions | 46.23% | 55.71% | 1.095 | 2.10x | $3,823 |
| Weekly | 46.61% | 55.31% | 1.102 | 1.74x | $2,974 |
| Monthly | 45.71% | 55.09% | 1.090 | 0.89x | $1,224 |
| Quarterly | 45.26% | 53.70% | 1.071 | 0.57x | $657 |

Daily and weekly 50/50 finish at almost identical annualized growth: 46.6152% and 46.6132%. The daily strategy nevertheless makes 2,884 fills and incurs $7,857 of modeled costs, versus 613 fills and $2,974 for weekly. The near equality of their net returns is specific to this price path.

The main-grid 50/50 CAGR ranges from 43.47% to 48.84% across recurring schedules. This is economically visible, but less than the allocation-driven endpoint range. Neither a weekly nor monthly label should be replaced mechanically by five or 21 sessions.

05 / INVESTOR EXPERIENCE

## 5 A high compound return can conceal a long recovery

*Daily closing drawdowns provide a clearer risk picture than terminal wealth alone.*

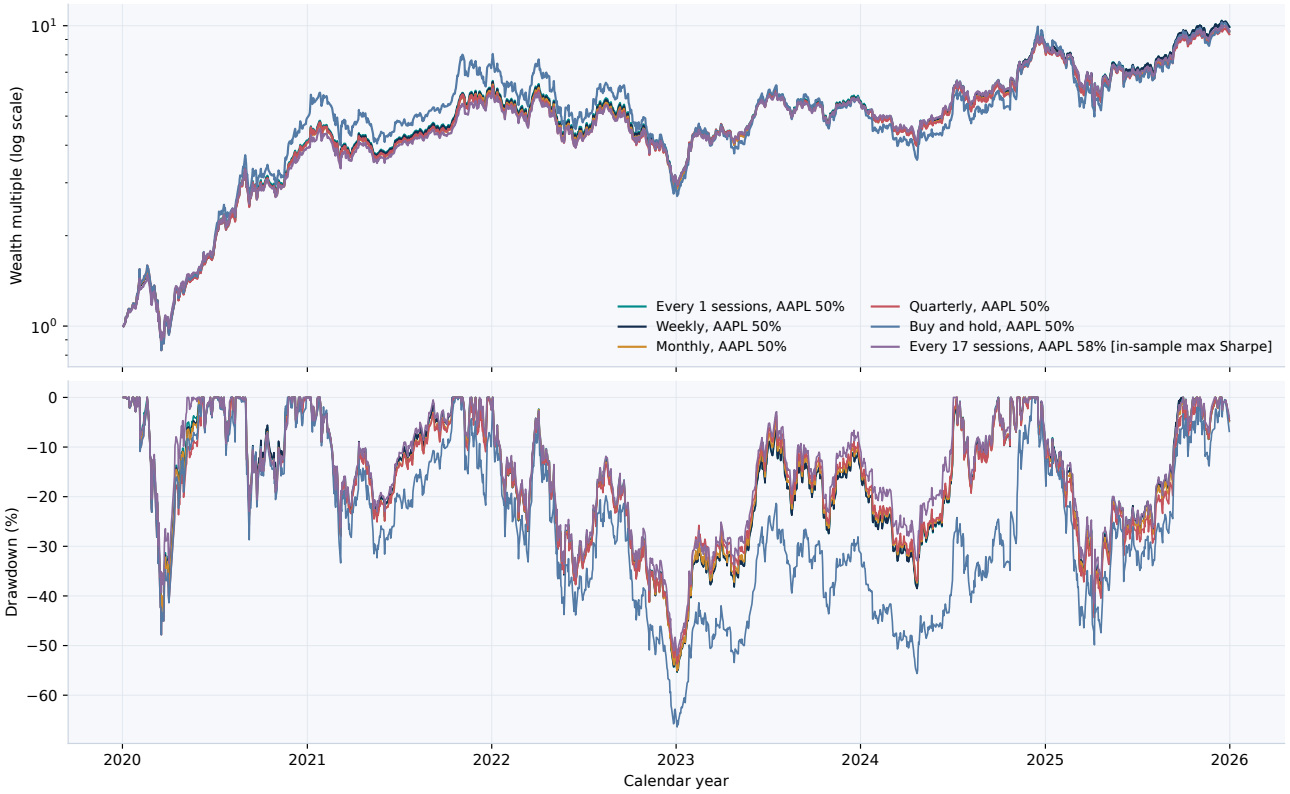



**Figure 4:** Top: net wealth multiples on a logarithmic scale. Bottom: drawdown from each strategy's running closing peak. The selected 58% AAPL / 17-session curve is labeled as an in-sample maximum Sharpe and is not a prespecified strategy.

| Portfolio | Max DD | Peak | Trough | Peak recovered |
|---|---|---|---|---|
| Weekly 50/50 | 55.31% | 2022-01-03 | 2023-01-03 | 2024-07-10 |
| 17 sessions, AAPL 58% | 52.02% | 2022-01-03 | 2023-01-03 | 2024-07-03 |
| Buy and hold, initial 50/50 | 66.41% | 2022-01-03 | 2023-01-03 | 2024-12-06 |

Weekly 50/50 reaches its deepest trough exactly one year after the recorded peak and takes more than two and a half years from that peak to recover. A strategy with impressive six-year growth may still require investors to tolerate a substantial and persistent loss.

Rebalancing improves the 50/50 drawdown relative to its buy-and-hold reference in this sample. It continually trims relative winners and adds to relative losers, so the benefit depends on the subsequent path of relative returns. Persistent relative trends can instead penalize rebalancing. Closing observations also miss intraday losses and do not capture the risk of execution during market stress.

06 / OBJECTIVES AND TRADEOFFS

# 6 There is no single best parameter pair

*Maximum wealth, maximum Sharpe and minimum drawdown answer different questions.*

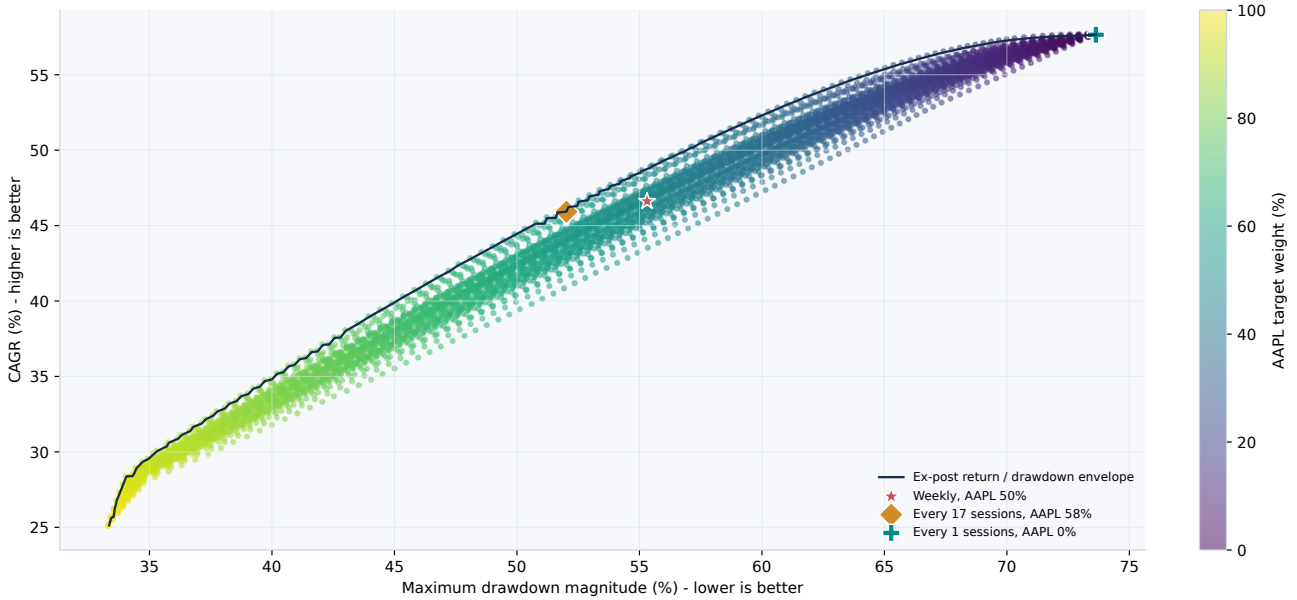



**Figure 5:** The point cloud includes recurring schedules; the envelope also considers buy and hold. Color denotes the AAPL target weight. The envelope is an ex-post upper return boundary at observed drawdown levels, not an estimated efficient frontier for future returns.

| Objective / reference | Schedule | AAPL | CAGR | Max DD | Sharpe |
|---|---|---|---|---|---|
| Maximum CAGR | 1 sessions | 0.00% | 57.65% | 73.63% | 1.017 |
| Maximum Sharpe | 17 sessions | 58.00% | 45.92% | 52.02% | 1.137 |
| Weekly reference | Weekly | 50.00% | 46.61% | 55.31% | 1.102 |
| 100% AAPL control | Weekly | 100.00% | 25.12% | 33.36% | 0.865 |

The maximum-CAGR result is the 100% TSLA endpoint, with 57.65% CAGR and a 73.63% drawdown. Because all schedules tie at this endpoint, naming a particular frequency as its winner would be misleading.

The maximum-Sharpe blend gives up growth to reduce volatility and drawdown. Its Sharpe improves by only 0.035 over weekly 50/50. No bootstrap confidence interval, multiple-testing adjustment or formal superiority test is claimed.

As an illustration of objective dependence, a full-sample search maximizing CAGR subject to observed drawdown below 40%, 50% or 60% selects AAPL weights of 81%, 60% and 39%, respectively. These thresholds are illustrative research constraints, not prospective loss limits. A future drawdown can exceed every observed bound.

07 / REGIME DEPENDENCE

# 7 Year-by-year results explain the aggregate

*The same target blend encounters very different market environments.*

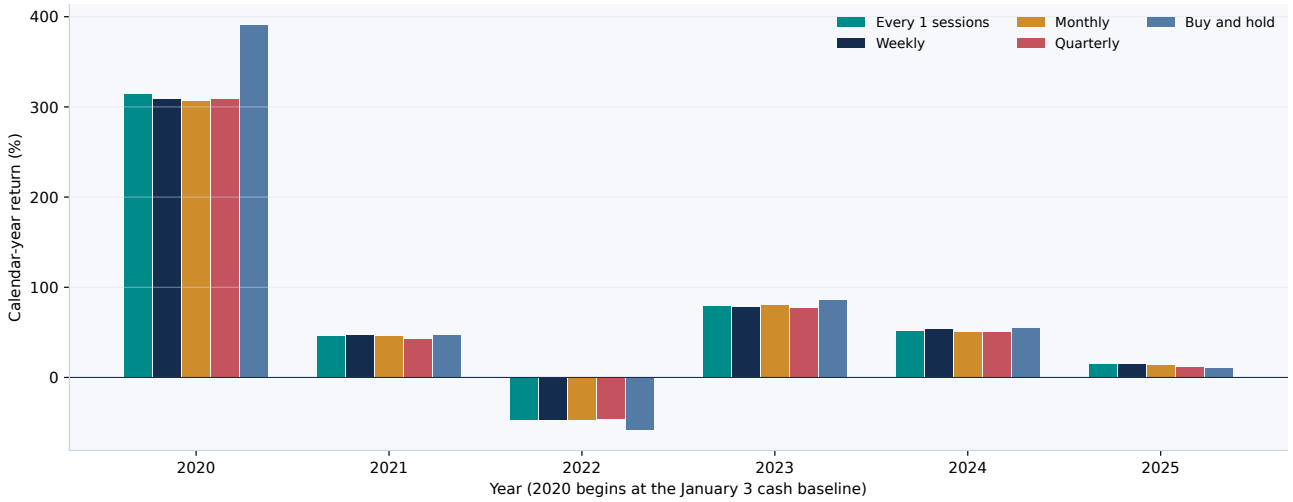



**Figure 6:** Calendar-year net returns of the fixed 50/50 references. The 2020 observation begins at the 3 January cash baseline. Subsequent years include the overnight return from the preceding year-end close.

| Year | Daily | Weekly | Monthly | Quarterly | Buy / hold |
|---|---|---|---|---|---|
| 2020 | 314.69% | 309.06% | 306.98% | 309.17% | 391.14% |
| 2021 | 46.37% | 47.18% | 45.51% | 42.59% | 46.95% |
| 2022 | -47.60% | -47.52% | -47.62% | -45.91% | -58.47% |
| 2023 | 79.04% | 78.34% | 80.47% | 77.10% | 85.84% |
| 2024 | 51.93% | 53.33% | 50.03% | 49.84% | 54.84% |
| 2025 | 14.51% | 14.65% | 13.65% | 11.89% | 10.89% |

The first year is unusually influential: weekly 50/50 returns 309.06% in 2020. Its 2022 return is −47.52%. The subsequent recovery does not erase the investor experience during that loss. Annualization compresses these very different episodes into one number.

Buy and hold benefits more from the 2020 run-up but loses more in 2022. Rebalancing repeatedly resets the portfolio toward equal value, limiting the accumulation of concentrated TSLA exposure. Its effect is therefore conditional on when large relative trends and reversals occur.

Annual returns are not independent replications of a six-year strategy test. The sample contains only six calendar years and a small number of major market regimes. These results cannot establish that the ranking will survive a new business cycle, a different start date or a different stock pair.

**Interpretation.** A large cumulative return does not establish an intrinsic "rebalancing premium." In this sample, market exposure, relative trends, weight drift and repeated trading all contribute.

08 / IMPLEMENTATION ROBUSTNESS

# 8 Costs and calendar phase can change the comparison

*A favorable anchor date should not be mistaken for a structural frequency effect.*

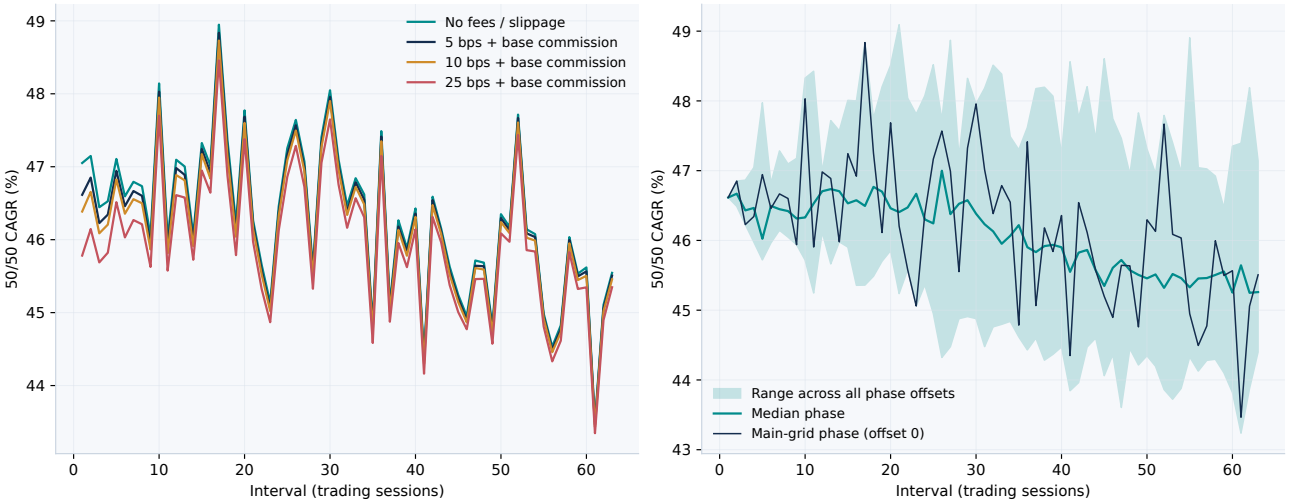



**Figure 7:** Left: full reruns with free execution or 5 / 10 / 25 bps adverse slippage plus base commission. Right: 2,016 fresh 50/50 simulations across every possible phase for each 1–63-session interval. All phase variants invest initially on 6 January 2020.

| Cost assumption | Daily CAGR | Weekly CAGR | Monthly CAGR | Quarterly CAGR |
|---|---|---|---|---|
| Free execution | 47.05% | 46.77% | 45.79% | 45.30% |
| 5 bps + fees | 46.62% | 46.61% | 45.71% | 45.26% |
| 10 bps + fees | 46.39% | 46.51% | 45.66% | 45.22% |
| 25 bps + fees | 45.78% | 46.16% | 45.45% | 45.11% |

At 50/50, daily execution loses about 0.44 percentage points of CAGR between free execution and the base-cost run, while weekly loses about 0.16 points. These are paired simulation differences including altered holdings and compounding; they are not just dollar fees divided by capital. Under 25 bps slippage plus fees, weekly exceeds daily CAGR by about 0.38 points.

For phase $p \in \{0, \ldots, k-1\}$, every strategy buys initially, then rebalances at session indices congruent to $p$ modulo $k$. Thus offset zero reproduces the main grid. The phase test changes rebalance timing without shifting the investment start date; some nonzero offsets have a short initial gap.

At $k = 5$, the 50/50 CAGR ranges from 45.72% to 47.97% across phases. At $k = 21$, it ranges from 45.24% to 49.09%. A 21-session interval at one phase can beat or trail another interval at another phase. The phase exercise is limited to 50/50 and does not validate the exact 58/42 winner across all phases.

09 / TEMPORAL VALIDATION

# 9 A training winner does not automatically beat a fixed rule

*Freeze the selected parameters before evaluating the later period.*

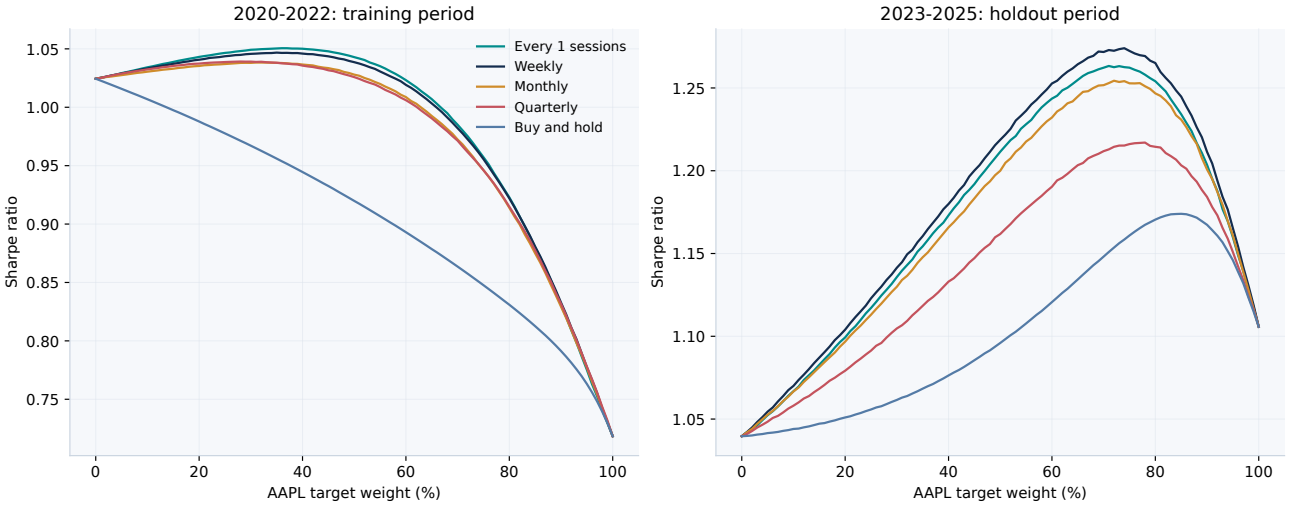



**Figure 8:** Sharpe versus target weight for fixed reference schedules. Training is 2020–2022; holdout is 2023–2025. Both start from fresh cash, include entry costs, and use their own preceding cash-only baseline. The holdout plot is descriptive; it is not used for training selection.

| Train objective | Schedule | AAPL | Train SR | Test CAGR | Test DD | Test SR |
|---|---|---|---|---|---|---|
| CAGR | 10 sessions | 1.00% | 1.027 | 55.68% | 53.56% | 1.042 |
| Sharpe | 17 sessions | 43.00% | 1.092 | 47.52% | 44.64% | 1.167 |
| Calmar | 17 sessions | 45.00% | 1.092 | 47.05% | 44.20% | 1.175 |
| Fixed weekly | Weekly | 50.00% | – | 47.09% | 43.14% | 1.219 |

Training ends at 30 December 2022 close. The later simulation uses that same date as a cash baseline and first buys at 3 January 2023 open, ending at 31 December 2025 close. Session intervals restart at that first purchase, so this is a new deployment of a selected policy, not a continuous portfolio carried across the split.

The training maximum-Sharpe rule selects 43% AAPL every 17 sessions. It delivers 47.52% holdout CAGR and a 1.167 Sharpe, versus 47.09% and 1.219 for fixed weekly 50/50. Its slightly greater CAGR comes with a deeper drawdown (44.64% versus 43.14%) and lower Sharpe. The training maximum-CAGR rule selects 1% AAPL, retaining a concentrated TSLA risk profile.

The apparent holdout is an internal chronological diagnostic, not a pristine external test: the two stocks and research question were chosen with knowledge of history, and the report displays the entire sample. There is one split, no nested selection process and no independent post-2025 evidence.

10 / RESEARCH CONCLUSION

# 10 Choose a robust policy before chasing a maximum

*The evidence supports a risk-budget decision; it does not identify a universal optimal frequency.*

### 1. Set the allocation from the loss you can tolerate

The AAPL/TSLA mix has a much larger broad effect on volatility and drawdown than the choice between ordinary weekly, monthly and quarterly schedules. Even 100% AAPL loses about one third from its worst closing peak in this sample. A two-stock long-only mandate cannot be assumed to deliver a small drawdown. Reducing the weight in TSLA reduces historical risk here, at the expense of historical growth.

### 2. Keep weekly 50/50 as a transparent reference

The original policy is competitive with daily, monthly and quarterly 50/50 in this period. It achieves approximately the daily strategy's base-cost CAGR with fewer than one quarter of its fills and substantially lower modeled trading costs. This supports its continued use as a research reference. It is not proof that weekly execution is optimal in future markets.

### 3. Treat 17 sessions and 58/42 as a candidate, not a conclusion

The full-sample Sharpe peak is an interior blend, but it is selected after observing 6,767 configurations. The chronological training winner chooses a different weight and does not beat weekly 50/50 on holdout Sharpe. Calendar-phase sensitivity also shows that frequency rankings can depend on the anchor. The exact full-sample optimum needs new, untouched data before it can justify a policy change.

### 4. Make the objective and execution contract explicit

Maximum CAGR favors TSLA concentration; minimum drawdown favors AAPL concentration; maximum Sharpe favors an interior allocation. A policy decision therefore requires an explicit priority among growth, loss tolerance, stability and implementation burden. None of these objectives provides a guarantee about future losses.

> **Practical reading of the evidence.** Weekly 50/50 is a reasonable baseline for this research program. The larger next decision is whether its observed 55% drawdown is acceptable. If it is not, changing the weight or broadening the permitted asset universe is a more direct research question than fine-tuning the rebalance day.

### Boundaries that matter

This is a retrospective two-stock study, subject to selection and survivorship bias. Prices and adjustments are vendor observations frozen in September 2026 rather than point-in-time vintage data. The study excludes taxes, auction uncertainty, market impact, liquidity capacity, borrow constraints, operational outages, financing and intraday drawdowns. The risk-free rate is zero, so the reported Sharpe is not an excess return over historical Treasury yields. Costs are a fixed stylized schedule, not an estimate from actual executions.

Future work should prespecify the allocation objective, freeze a small candidate set before collecting new observations, and test any proposed execution rule on untouched data. Expanding to other stocks, introducing threshold bands, tax-aware turnover or dynamic signals would be new research with different assumptions; none is claimed completed here.

11 / DEFINITIONS

# 11 Metrics and accounting conventions

*Every headline statistic is computed from the same net daily equity path.*

Let $E_0 = 100{,}000$ be the cash baseline, $E_t$ net closing equity, $N = 1506$ observed returns and $Y = (d_N - d_0)/365.25$. Define $r_t = E_t/E_{t-1} - 1$ for $t = 1, \ldots, N$.

$$R_{\text{total}} = E_N/E_0 - 1, \qquad \text{CAGR} = (E_N/E_0)^{1/Y} - 1,$$
$$\sigma_{\text{ann}} = \sqrt{252}\, s(r_t), \qquad \text{Sharpe} = \sqrt{252}\, \bar{r}/s(r_t),$$
$$D_t = E_t/\max_{0\le j\le t} E_j - 1, \qquad \text{MDD} = -\min_t D_t,$$
$$\text{Calmar} = \text{CAGR}/\text{MDD}\,.$$

The sample standard deviation uses denominator $N - 1$. Sharpe uses arithmetic daily mean return and a zero risk-free rate; CAGR is geometric. They answer different questions. MDD is presented as a positive loss magnitude throughout the report. Drawdown curves themselves use the negative signed $D_t$.

### Execution and costs

At each scheduled open, the pre-trade equity is cash plus holdings marked at adjusted opening prices. The broker freezes that equity for both target calculations. A reference target $\lfloor w_i E/O_i \rfloor$ determines whether the instrument needs to be bought or sold. The executable integer target then uses the adverse fill price $O_i(1 \pm b/10{,}000)$. Only quantities consistent with the initial trade direction are submitted. Sells are completed before cash-constrained buys.

Commission per executed order is $\max(1, 0.005|q|)$ USD. Slippage attribution is $|q|\,|P_{\text{fill}} - O|$, already reflected in the fill and not deducted a second time. Reported total cost is the dollar sum of commission and slippage over the sample, with no annualization. Free execution removes both charges; the 10 and 25 bps cases retain the base commission.

### Turnover

The report uses annual two-sided gross traded opening notional:

$$T_{\text{ann}} = \frac{1}{Y} \sum_{\text{fills } j} \frac{|q_j| O_{i(j),t(j)}}{E^{\text{pre-trade open}}_{t(j)}}.$$

The initial purchase is included. This diagnostic uses opening reference notional and elapsed calendar years. It intentionally differs from QuantLab's archived turnover metric, which uses fill notional divided by prior closing equity and scales by $252/N$. Equity, returns, costs, Sharpe and drawdown retain the original definitions; turnover figures across these two conventions should not be compared as identical.

### Buy-and-hold and year boundaries

Buy and hold makes only the initial allocation and allows weights to drift. Yearly returns use consecutive year-end equity, except that 2020 starts at the original cash baseline. Terminal wealth includes marked positions, with no final liquidation. The chronological split uses fresh cash and a reset session anchor in the test period; its metrics cannot be compounded into the original continuous strategy without changing the investment rule.

12 / AUDIT AND REPRODUCIBILITY

# 12 Frozen inputs, verified calculations and sources

*Inputs, research code and numerical outputs are saved separately for inspection and offline reproduction.*

### Input provenance and validation

Source: Yahoo Finance via yfinance 0.2.66, downloaded on 18 September 2026 at 02:12:25 UTC, with `auto_adjust=False` and `actions=True`. The source archive contains 4,521 rows for AAPL, TSLA and SPY; only AAPL and TSLA enter the portfolio calculations. The main study deliberately preserves the original 2020–2025 horizon.

All 1,507 dates match the XNYS exchange-session calendar. Checks find no duplicate date/symbol keys, missing OHLCV observations, nonpositive prices or volumes, or OHLC range inconsistencies for the three archived instruments. These are internal consistency checks, not an independent second-vendor price reconciliation. The archived weekly snapshot matches the original input frame exactly.

**SHA-256 of the source Parquet file:**
`9ea93af1777f51df9631178c2f029efcaf43958f35a9ba7538f8f173c91b9348`

### Calculation verification

The vectorized sweep reproduces every archived weekly 50/50 closing equity observation; the maximum absolute difference is $3.50 \times 10^{-10}$ USD. Six additional full-path comparisons against the original QuantLab broker cover daily 100% TSLA, every three sessions at 37% AAPL, weekly 50/50, monthly 100% AAPL, quarterly 20% AAPL and buy-and-hold 50/50. Each comparison has zero observed difference in double-precision equity. Cash and holdings are checked for nonnegativity; all reported core metrics are finite. The maximum immediate post-rebalance target-weight mismatch in the base grid is 0.209 percentage points.

### Reproduction files

The project keeps the report source separate from its numerical record. `data/` contains the frozen market-data Parquet, provider metadata and original weekly-reference equity. `results/` contains all base, cost, training, holdout and phase grids as CSV, full base and holdout equity paths in a compressed NumPy archive, and the calculation audit summary.

`code/research.py` runs the simulations, checks broker parity, calculates all reported metrics and renders all eight figures. `code/build_report.py` regenerates this TeX document from the saved results. `code/quantlab_reference/` freezes the original broker, strategy and configuration used for verification. The observed environment versions and input/output SHA-256 hashes are recorded in `results/`.

From the project directory, using a Python environment with the supplied dependencies, run:

```
python code/research.py
python code/build_report.py
pdflatex -output-directory=/tmp/quantlab-build rebalancing_study.tex
```

Create the build directory first. Compile twice to refresh references, then copy the resulting PDF into the project directory. An optional `QUANTLAB_REPORT_DIR` or `QUANTLAB_BUILD_DIR` redirects figures or numerical outputs during a verification run. No network request is required. The original dependency lock file and a short reproduction guide accompany the code.

### Source references

[1] Yahoo Help, *What is the adjusted close?* Describes adjustment for applicable splits and dividend distributions. https://in.help.yahoo.com/kb/adjusted-close-sln28256.html

[2] Tesla Investor Relations, *Tesla Announces a Three-for-One Stock Split*, 5 August 2022. Corporate-action context; split adjustment is handled by the vendor price series. https://ir.tesla.com/press-release/tesla-announces-three-one-stock-split

[3] QuantLab local research archive, weekly equal-weight run `20260918_021504_weekly_equal_weight_aapl_tsla_f328c3`; frozen source and reference equity included in the project data and code directories.